Practice 2 Report: Validation and Decision Trees

In this Lab I looked at how reliable the evaluation metrics are for machine learning depending on how we validate things. Focusing on Decision Trees and using two main methods: Hold-out and 10-fold Cross-Validation using the scikit-learn library.

I ran 20 separate tests using the given dataset:

- **Hold-out Method:** 10 tests with a totally random 90/10 split for training and testing.
- **Cross-Validation Method:** 10 tests using a 10-fold CV setup, also with randomized parts.
- I made sure not to use any random seed so that every time the partitions were different.

Hold-out Results

Exp	Time (s)	Accuracy	Recall	Precision	TNR	F1-Score	AUC-ROC
1	0.0259	0.58	0.50	0.80	0.75	0.62	0.62
2	0.0233	0.92	0.90	1.00	1.00	0.95	0.95
3	0.0230	0.83	1.00	0.71	0.71	0.83	0.86
4	0.0224	0.67	0.56	1.00	1.00	0.71	0.78
5	0.0256	0.83	0.67	1.00	1.00	0.80	0.83

6	0.0247	0.67	0.86	0.67	0.40	0.75	0.63
7	0.0238	0.58	0.71	0.62	0.40	0.67	0.56
8	0.0280	0.67	0.57	0.80	0.80	0.67	0.69
9	0.0445	0.75	0.50	1.00	1.00	0.67	0.75
10	0.0495	0.67	0.86	0.67	0.40	0.75	0.63

10-Fold Cross-Validation Results

Exp	Time (s)	Accuracy	Recall	Precision	TNR	F1-Score	AUC-ROC
1	0.2645	0.66	0.72	0.72	0.68	0.69	0.70
2	0.1801	0.71	0.72	0.76	0.74	0.72	0.73
3	0.1846	0.70	0.73	0.72	0.66	0.72	0.70

4	0.1737	0.73	0.79	0.76	0.68	0.76	0.74
5	0.1884	0.70	0.76	0.74	0.66	0.73	0.71
6	0.1732	0.72	0.74	0.77	0.67	0.74	0.71
7	0.1753	0.70	0.72	0.77	0.68	0.73	0.70
8	0.1704	0.66	0.69	0.75	0.69	0.69	0.69
9	0.1927	0.68	0.74	0.71	0.62	0.71	0.68
10	0.1741	0.64	0.72	0.71	0.59	0.69	0.66

Results Comparison

The numbers show a big difference in how much we can trust the results.

- **Huge Variance in Hold-out:** Test 2 got a very high Accuracy of 0.92, but Test 7 dropped down to 0.58 even though the same setup was used. This happens because the dataset is very small, with only 120 rows. A 10% test size means the model is only tested on 12 samples. The numbers aren't really showing what the model can do. It's just “luck” because we only used a tiny test set. The model is “too good” for the test data, so it gets confused and gives random scores.
- **Stability of Cross-Validation:** Here it’s a bit better, it checks all the data 10 times, so it averages out the bad results. It's more stable because it gets to know the dataset completely, making the

final score much more honest and accurate. The Accuracy stays in a small range between 0.64 and 0.73 across 10 random experiments, which is much more believable.

- **Computational Trade-off:** Cross validation takes way longer, 0.18 seconds, which is a massive delay compared to the 0.025 seconds for Hold-out. All due to the computer having to run the whole training process over and over, ten full times, which is obviously slower.

Conclusions

This practice proves Hold-out is quite inefficient for tiny data sets. We can see how we can't trust the numbers at all. It's only good if we need a quick job. Making 10-fold Cross-Validation the only viable choice to get scores that are close to the actual truth even if it takes longer.

References

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-Hold-out
Exp 1 | Tiempo: 0.0343s | Acc: 1.00 | Rec: 1.00 | Prec: 1.00 | TNR: 1.00 | F1: 1.00 | AUC: 1.00
Exp 2 | Tiempo: 0.0229s | Acc: 0.75 | Rec: 0.71 | Prec: 0.83 | TNR: 0.80 | F1: 0.77 | AUC: 0.76
Exp 3 | Tiempo: 0.0213s | Acc: 0.75 | Rec: 0.71 | Prec: 0.83 | TNR: 0.80 | F1: 0.77 | AUC: 0.76
Exp 4 | Tiempo: 0.0220s | Acc: 0.50 | Rec: 0.38 | Prec: 0.75 | TNR: 0.75 | F1: 0.50 | AUC: 0.56
Exp 5 | Tiempo: 0.0217s | Acc: 0.50 | Rec: 0.67 | Prec: 0.29 | TNR: 0.44 | F1: 0.40 | AUC: 0.56
Exp 6 | Tiempo: 0.0226s | Acc: 0.75 | Rec: 1.00 | Prec: 0.70 | TNR: 0.40 | F1: 0.82 | AUC: 0.70
Exp 7 | Tiempo: 0.0218s | Acc: 0.67 | Rec: 0.62 | Prec: 0.83 | TNR: 0.75 | F1: 0.71 | AUC: 0.69
Exp 8 | Tiempo: 0.0202s | Acc: 0.75 | Rec: 0.67 | Prec: 1.00 | TNR: 1.00 | F1: 0.80 | AUC: 0.83
Exp 9 | Tiempo: 0.0220s | Acc: 0.83 | Rec: 0.80 | Prec: 1.00 | TNR: 1.00 | F1: 0.89 | AUC: 0.90
Exp 10 | Tiempo: 0.0329s | Acc: 0.58 | Rec: 0.50 | Prec: 0.60 | TNR: 0.67 | F1: 0.55 | AUC: 0.58

10-Fold Cross-Validation
Exp 1 | Time: 0.6283s | Acc: 0.69 | Rec: 0.71 | Prec: 0.75 | TNR: 0.67 | F1: 0.72 | AUC: 0.69
Exp 2 | Time: 0.3667s | Acc: 0.68 | Rec: 0.74 | Prec: 0.75 | TNR: 0.66 | F1: 0.72 | AUC: 0.70
Exp 3 | Time: 0.3787s | Acc: 0.71 | Rec: 0.71 | Prec: 0.76 | TNR: 0.73 | F1: 0.70 | AUC: 0.72
Exp 4 | Time: 0.2827s | Acc: 0.70 | Rec: 0.72 | Prec: 0.75 | TNR: 0.67 | F1: 0.71 | AUC: 0.70
Exp 5 | Time: 0.4832s | Acc: 0.69 | Rec: 0.72 | Prec: 0.72 | TNR: 0.61 | F1: 0.70 | AUC: 0.67
Exp 6 | Time: 0.3880s | Acc: 0.69 | Rec: 0.73 | Prec: 0.76 | TNR: 0.67 | F1: 0.72 | AUC: 0.70
Exp 7 | Time: 0.3302s | Acc: 0.68 | Rec: 0.72 | Prec: 0.73 | TNR: 0.64 | F1: 0.71 | AUC: 0.68
Exp 8 | Time: 0.2796s | Acc: 0.72 | Rec: 0.78 | Prec: 0.74 | TNR: 0.62 | F1: 0.75 | AUC: 0.70
Exp 9 | Time: 0.2591s | Acc: 0.72 | Rec: 0.80 | Prec: 0.75 | TNR: 0.64 | F1: 0.77 | AUC: 0.72
Exp 10 | Time: 0.2300s | Acc: 0.71 | Rec: 0.72 | Prec: 0.71 | TNR: 0.67 | F1: 0.72 | AUC: 0.70
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